

Advantages of Plane Table Surveying

1. Field observation and plotting are done simultaneously.
 - The map is drawn directly in the field while observing the ground.
2. Less chance of missing details.
 - Since the surveyor sees the actual features while plotting, important objects are rarely omitted.
3. Errors can be detected and corrected immediately.
 - The plotted map can be compared with the actual ground features on the spot.
4. Fast and economical.
 - No separate office plotting is required, saving both time and cost.
5. Suitable for small-scale surveys.
 - Best for topographical surveys, route surveys, and reconnaissance surveys.
6. Less field book work.
 - Measurements are plotted directly, so very few notes are needed.
7. Irregular features can be plotted easily.
 - Rivers, roads, ponds, and buildings can be sketched directly.

Disadvantages of Plane Table Surveying

1. **Lower accuracy than other surveying methods.**
 - Not suitable where high precision is required.
2. **Cannot be used in bad weather.**
 - Rain, strong wind, or poor visibility affects the work.
3. **Difficult in dense forests or hilly areas.**
 - Clear visibility between stations is necessary.
4. **Large equipment is inconvenient.**
 - The table and tripod are bulky and difficult to transport.
5. **Paper may expand or shrink.**
 - Moisture and temperature changes can affect the accuracy of the map.
6. **Not suitable for large-area surveys.**
 - More accurate methods like theodolite or total station are preferred.
7. **Requires proper orientation at every station.**
 - Incorrect orientation causes plotting errors throughout the survey.

Key Principle

- Plane table surveying is based on the **principle of parallelism**.
- **Definition:** *All rays (lines) drawn from the survey station on the map must remain parallel to the corresponding lines on the ground.*
- At every survey station, the plane table must be **properly oriented** so that it has the same direction (usually towards **magnetic north**) as at the previous station.

three essential processes:

1. **Leveling**
2. **Centering**
3. **Orientation**

1. Leveling

Definition

Leveling is the process of making the **plane table perfectly horizontal** so that accurate plotting can be done.

Procedure

- Set the plane table at a convenient working height (about **1 m**).
- Spread the tripod legs well apart and fix them firmly in the ground.
- Place a **spirit level** on the board.
- Check the level in **two perpendicular (90°) directions**.
- Adjust the tripod until the bubble remains at the center in both directions.

Purpose

- Keeps the drawing sheet horizontal.
- Ensures accurate plotting and measurement.
- Prevents errors due to a tilted board.

Instrument Used: Spirit Level

2. Centering

Definition

Centering is the process of placing the plane table so that the **plotted station point on the drawing sheet lies exactly above the corresponding ground station.**

Procedure

- Move the table until the plotted station point is directly over the survey station.
- Use a **U-fork (Plumbing Fork)** and **Plumb Bob** to check the exact position.
- Adjust the table until perfect alignment is achieved.

Purpose

- Ensures the plotted point represents the exact ground position.
- Reduces plotting errors.

Instrument Used: U-fork (Plumbing Fork) and Plumb Bob

3. Orientation

Definition

Orientation is the process of keeping the **plane table in the same direction (parallel position)** at every survey station.

Principle

When properly oriented,

- **Lines drawn on the paper remain parallel to the corresponding lines on the ground.**
- This satisfies the **Principle of Parallelism.**

Purpose

- Maintains the correct direction of all plotted lines.
 - Ensures continuity and accuracy of the survey.
-

Methods of Orientation

There are **two methods**:

1. By Magnetic Needle
 2. By Back Sighting
-

A. Orientation by Magnetic Needle

Procedure

- At the first station, draw the **magnetic north line** on the drawing sheet.
- At the next station, place the **trough compass** along the north line.
- Rotate the table until the magnetic needle points to **0°–0°** on the compass.
- Clamp the table in this position.

Advantages

- Simple and quick.
- Easy for beginners.

Disadvantages

- Less accurate.
- Affected by **local attraction** (nearby magnetic objects like iron, steel, electric lines).

Instrument Used: Trough Compass

B. Orientation by Back Sighting

Procedure

Suppose:

- **A = First survey station**
 - **B = Second survey station**
1. At station **A**, draw line **ab** representing ground line **AB**.
 2. Move the table to station **B**.
 3. Rotate the table until line **ba** points exactly toward the ranging rod placed at station **A**.
 4. Clamp the table.

Advantages

- More accurate than the magnetic needle method.
- Not affected by local magnetic attraction.
- Gives **perfect orientation.**

Disadvantages : Requires visibility between two stations.

Magnetic Needle	Back Sighting
Uses trough compass	Uses previous survey station
Quick and simple	More accurate
Affected by local attraction	Not affected by local attraction
Less accurate	Gives perfect orientation
Suitable for rough surveys	Suitable for precise surveys

Plane table surveying can be carried out by **four methods**:

1. **Radiation Method**
2. **Intersection Method**
3. **Traversing Method**
4. **Resection Method**

1. Radiation Method

Definition

In the **Radiation Method**, all objects are plotted by drawing **rays from a single survey station**. The distance from the station to each object is measured and plotted to scale.

Principle

- **One station → Many objects**
- Direction is obtained by the **Alidade**.
- Distance is measured on the ground and plotted on the sheet.

Procedure

1. Set up the plane table at station **P**.
2. Level, center and orient the table.
3. Mark point **p** on the sheet representing station **P**.
4. Draw the north line.
5. Place the alidade on point **p**.
6. Sight objects **A, B, C and D** and draw rays.
7. Measure distances **PA, PB, PC and PD**.
8. Plot these distances to scale on the respective rays to obtain points **a, b, c and d**.

Advantages

- Very simple and quick.
- Suitable for small areas.
- Requires only one survey station.

Disadvantages

- Accuracy decreases for distant objects.
- Not suitable where distance measurement is difficult.

Suitable For

- Small open areas.
- Topographical surveys.
- Objects clearly visible from one station.

2. Intersection Method

Definition

In the **Intersection Method**, the position of an object is obtained by the **intersection of rays drawn from two different survey stations**.

Principle

- **Two stations → One object**
- Distance to the object is **not measured**.

Procedure

Suppose **A** and **B** are survey stations and **P** is the object.

1. Set up the table at **A**.
2. Level, center and orient the table.
3. Draw rays towards **B** and **P**.

4. Measure distance **AB** and plot point **b**.
 5. Shift the table to station **B**.
 6. Orient the table by **Back Sighting**.
 7. Draw another ray towards **P**.
 8. The intersection of the two rays gives point **p**, the location of object **P**.
-

Advantages

- Distance to the object is not required.
 - Suitable for inaccessible objects.
 - More accurate for long distances.
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Disadvantages

- Requires accurate orientation.
 - Small orientation errors cause large plotting errors.
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Suitable For

- Rivers
 - Lakes
 - Mountains
 - Valleys
 - Objects that cannot be reached
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3. Traversing Method

Definition

The **Traversing Method** is used to survey an area by connecting a **series of survey stations**.

Principle

Stations are connected one after another.

Example:

A → B → C → D

Procedure

1. Select traverse stations (A, B, C, D).
 2. Set up the table at station **A**.
 3. Level, center and orient the table.
 4. Draw a ray towards station **B**.
 5. Measure distance **AB** and plot it.
 6. Shift the table to **B**.
 7. Orient by **Back Sighting**.
 8. Draw ray towards **C**.
 9. Measure and plot **BC**.
 10. Repeat until all stations are connected.
-

Check of Accuracy

Check Line

- Sight back to previous stations.
- Used to verify plotting accuracy.

Error of Closure

- In a closed traverse, the final station should coincide with the first station.
 - Any difference is called the **Error of Closure**.
 - If the error is within the permissible limit, it is adjusted graphically.
-

Advantages

- Suitable for long narrow areas.
 - Good accuracy.
 - Can cover large areas.
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Disadvantages

- Time-consuming.
 - Requires careful measurements and orientation.
-

Suitable For

- Roads
- Railways
- Canals
- Pipelines
- Property boundaries

4. Resection Method

Definition

Resection is the process of determining the **position of the plane table station** by taking sights to **known points whose positions are already plotted**.

Principle

- Known points → Unknown station.

Methods of Resection

There are **four methods**:

1. By Compass
2. By Back Sighting
3. By Two-Point Problem
4. By Three-Point Problem

Procedure (General)

Suppose **A** and **B** are known stations and **P** is the unknown station.

1. Measure and plot baseline **AB**.
2. Set up the table at station **A**.
3. Orient the table by sighting **B**.
4. Draw a ray towards **P**.
5. Shift the table near station **P**.
6. Orient by **Back Sighting**.
7. Draw a ray from **B**.
8. The intersection gives the plotted position **P**.
9. Mark the actual ground station using a **U-fork and Plumb Bob**.

Advantages

- No need to establish the station beforehand.
- Useful when the station position is unknown.
- Saves time in fieldwork.

Disadvantages

- Requires previously plotted reference points.
- Needs accurate orientation.

Suitable For

- Extending existing surveys.
- Locating new survey stations.
- Survey continuation.

Comparison

Method	Main Principle	Distance Measured?	Best Used For
Radiation	Rays from one station	✓ Yes	Small open areas
Intersection	Intersection of rays from two stations	✗ No	Rivers, mountains, inaccessible objects
Traversing	Series of connected stations	✓ Yes	Roads, canals, railways, boundaries
Resection	Find unknown station from known points	Usually No	Extending existing surveys

METHODS OF PLANE TABLING

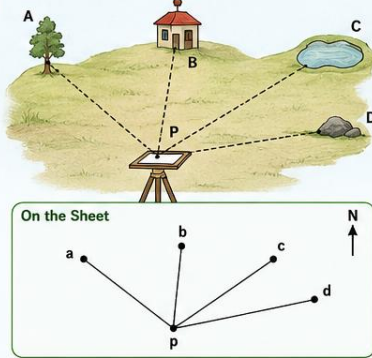
1. RADIATION METHOD

Principle :

All objects are plotted by drawing rays from a single survey station.

Procedure :

1. Set up the table at station P.
2. Level, center and orient the table.
3. Mark point p on the sheet.
4. Sight objects A, B, C and D and draw rays.
5. Measure PA, PB, PC and PD.
6. Plot these distances to scale on the rays to get points a, b, c and d.



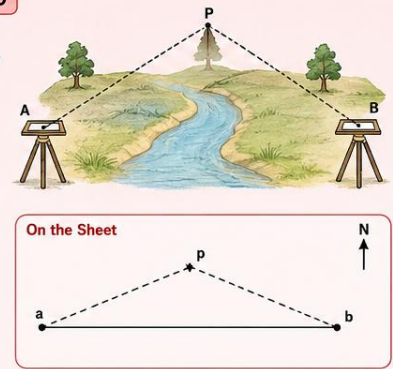
2. INTERSECTION METHOD

Principle :

The position of an object is obtained by the intersection of rays drawn from two different stations.

Procedure :

1. Set up the table at station A.
2. Draw rays towards B and object P.
3. Measure AB and plot point b.
4. Shift the table to station B and orient by back sighting A.
5. Draw a ray towards P.
6. Intersection of the two rays gives point p (position of P).



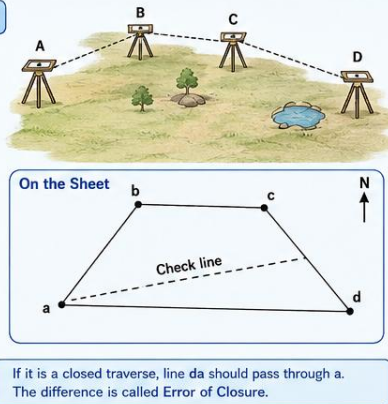
3. TRAVERSING METHOD

Principle :

A series of stations are connected one after another.

Procedure :

1. Select traverse stations A, B, C, D.
2. Set up table at A, orient and draw ray to B.
3. Measure AB and plot it.
4. Shift table to B, orient by back sighting A.
5. Draw ray to C, measure BC and plot it.
6. Repeat the same for C to D.
7. Check by taking check lines.
8. In closed traverse, any closing error is adjusted graphically.



4. RESECTION METHOD

Principle :

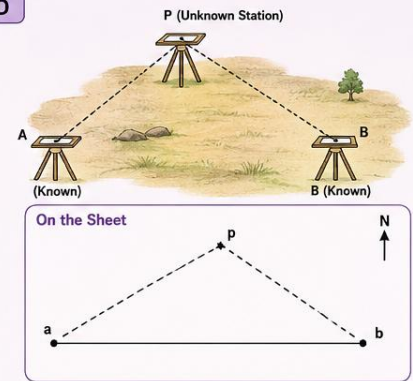
The position of an unknown station is determined from known plotted points.

Methods of Resection :

1. By Compass
2. By Back Sighting
3. By Two Point Problem
4. By Three Point Problem

Procedure (Two Point Problem) :

1. Measure and plot baseline AB.
2. Set up table at A and orient by sighting B.
3. Draw ray towards unknown station P.
4. Shift table near P and orient by back sighting A.
5. Draw ray from B.
6. Intersection gives position of P.
7. Mark the point on ground by U-fork and plumb-bob.



Symbols : • A, B, C, D = Survey Stations • P = Object / Unknown Station • a, b, c, d = Plotted Points

In all methods : Table must be Leveled, Centered and Oriented properly.

Methods of Resection

There are **4 methods**:

1. By Compass
2. By Back Sighting
3. Two-Point Problem
4. Three-Point Problem

1. By Compass Method

Definition

The position of the new station is determined by **orienting the table with a compass** and drawing rays from known points.

Procedure

1. Set up the plane table at the unknown station C.
2. Orient the table using a **compass**.
3. Clamp the table.
4. Draw rays from plotted points a and b towards ground stations A and B.
5. The intersection of the rays gives point c, the position of station C.

Advantages

- Simple and quick.

Disadvantages

- Less accurate.
- Affected by local attraction.

Suitable For

- Small-scale or rough surveys.

2. By Back Sighting

Definition

The table is oriented by sighting a previously occupied station.

Procedure

1. Shift the table to the new station.
2. Center it properly.
3. Orient by **back sighting** to the previous station.
4. Draw a ray towards another known point.
5. The intersection of rays gives the required station.

Advantages

- More accurate than compass method.
 - Not affected by magnetic attraction.
-

3. Two-Point Problem

Definition

The position of a new station is determined using **two known plotted points**.

Principle

- **2 known points → Find unknown station**
 - Uses **one auxiliary station**.
-

Procedure

1. Select two known points **P** and **Q** (plotted as **p** and **q**).
 2. Choose an auxiliary station **B**.
 3. Set up the table at **B**.
 4. Draw rays towards **P** and **Q**.
 5. Their intersection gives point **b**.
 6. Draw a ray towards the required station **A**.
 7. Shift the table to **A**.
 8. Orient by back sighting.
 9. Draw rays towards **P** and **Q** again.
 10. Remove the **triangle of error**.
 11. Final intersection gives point **a**, the correct station.
-

Triangle of Error

Definition

When the rays do **not meet at one point** because of imperfect orientation, they form a **small triangle**, called the **Triangle of Error**.

Elimination

- Rotate the table slightly.
 - Repeat orientation until the triangle disappears.
-

Advantages

- Good accuracy.
- Useful when only two known stations are available.

Disadvantages

- Auxiliary station is required.
 - Slightly time-consuming.
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4. Three-Point Problem

Definition

The position of an unknown station is determined using **three known plotted points**.

Principle

- **3 known points → Find unknown station**
 - **No auxiliary station is required.**
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Methods of Solving Three-Point Problem

(a) Bessel's Method (Graphical Method)

Procedure

1. Set up the table at unknown station **P**.
2. Level the table.
3. Sight known points **A, B and C**.
4. Draw rays according to Bessel's procedure.
5. Orient correctly.
6. Final rays intersect at point **p**.
7. Transfer point **p** to the ground using **U-fork and Plumb Bob**.

Advantages

- Very accurate.
- Most commonly used graphical method.

(b) Mechanical Method

Procedure

1. Place tracing paper over the map.
2. Mark a temporary point.
3. Draw rays towards **A, B and C**.
4. Move the tracing paper until all rays pass through **a, b and c** simultaneously.
5. Prick the correct point on the map.
6. Remove tracing paper.
7. Draw final rays.

Advantages

- Quick.
- Easy graphical solution.

(c) Trial and Error Method

Procedure

1. Set up the table at the required station.
2. Orient approximately by eye.
3. Draw rays towards **A, B and C**.
4. A small **triangle of error** will usually form.
5. Rotate the table clockwise or anticlockwise repeatedly.
6. Continue until all three rays meet at one point.
7. That point is the required station.

Advantages

- Simple.
- No special construction required.

Disadvantages

- Time-consuming.
- Depends on the surveyor's experience.

Method	Uses	Accuracy	Special Feature
Compass	Compass	Low	Used for rough surveys
Back Sighting	Previous station	High	Preferred method
Two-Point Problem	Two known points	High	Requires auxiliary station
Three-Point Problem	Three known points	Very High	No auxiliary station required

Errors in Plane Table Surveying

1. Instrumental Errors

These occur because of defects in the surveying instruments.

Causes

- Drawing board surface is not flat.
- Edge of the alidade is not straight.
- Object vane and sight vane are not perpendicular.
- Edge of alidade is not parallel to the line of sight.
- Defective fixing clamp.

- Defective sighting.
- Defective orientation.
- Defective centering.
- Movement of the board between sightings.

3. Errors in Plotting

- These occur while drawing on the sheet. Wrong map scale.

Causes

- Wrong map scale.
- Incorrect intersection of rays from different stations.

2. Errors in Manipulation and Sighting

These occur due to mistakes made by the surveyor.

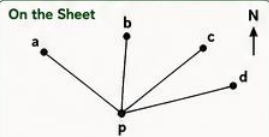
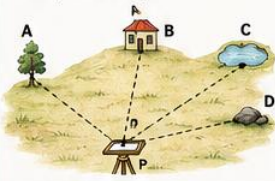
- Defective leveling.

PLANE TABLE SURVEYING – METHODS, RESECTION & ERRORS

METHODS OF PLANE TABLING

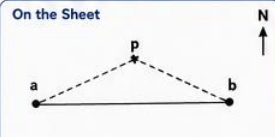
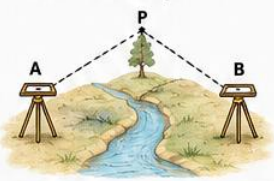
1. RADIATION METHOD

- One station – many objects.
- Distances are measured and plotted on rays.



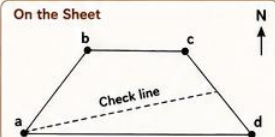
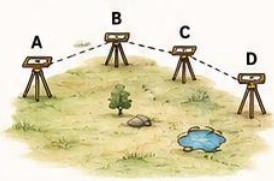
2. INTERSECTION METHOD

- Two stations – one object.
- Position of object is obtained by intersection of two rays.



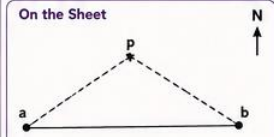
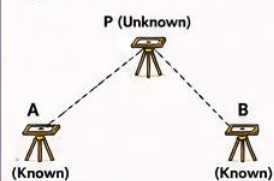
3. TRAVERSING METHOD

- Series of connected stations.
- Distances between stations are measured and plotted.



4. RESECTION METHOD

- Position of an unknown station
- is determined from known points.



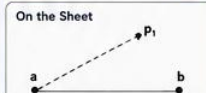
METHOD OF RESECTION

GENERAL IDEA

- The position of unknown station P is found from already plotted known points.
- After resection, the station is marked on ground by U-fork and plumb-bob.

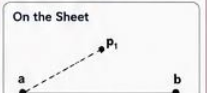
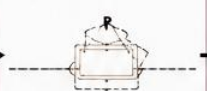
1. At station A

- Orient table towards B.
- Sight P and draw ray.



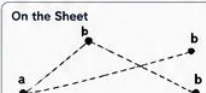
2. Shift table near P

- Center so that P1 is just over P.



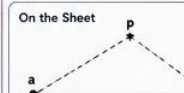
3. Orient by back sighting A

- With alidade on b, sight B and draw ray.



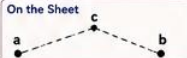
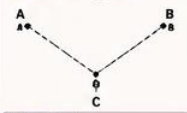
4. Intersection gives P

- Intersection of two rays gives plotted position P.
- Mark P on ground by U-fork and plumb-bob.



A. BY COMPASS

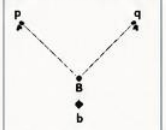
- Used only for small scale or rough mapping.
- Orient table with compass.
- From a and b draw rays towards A and B.
- Intersection gives c.



B. TWO POINT PROBLEM

- Select two points P and Q (plotted as p and q).
- Take an auxiliary station B.
- From B, rays to P and Q meet at b.
- Sight to required station A and mark a1.
- Shift table to A, back sight and draw ray to P.
- Correct orientation by eliminating triangle of error.
- Final intersection gives a (position of A).

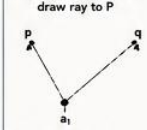
(i) From auxiliary station B



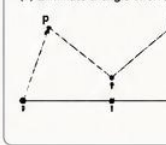
(ii) Sight A and mark a1



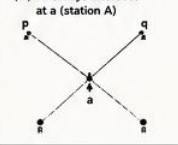
(iii) At A, back sight & draw ray to P



(v) Eliminate triangle of error



(vi) Final rays intersect at a (station A)

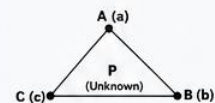


C. THREE POINT PROBLEM

Use three known points A, B, C (plotted as a, b, c).

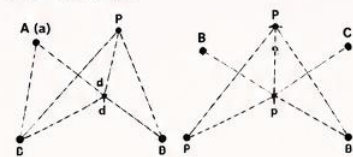
Methods:

- Bessel's Method
- Mechanical Method
- Trial and Error Method



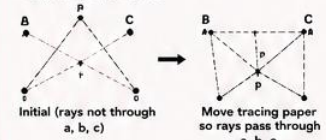
(i) BESSEL'S METHOD (GRAPHICAL)

- Place table at P and level.
- Sight A (along ca), then B (from c) → draw ray.
- Sight C (along ac), then B (from a) → draw ray.
- Rays meet at d.
- Sight B along db to orient.
- Draw final rays to A, B, C → meet at p.
- Mark p on ground.



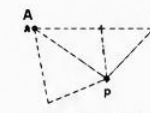
(ii) MECHANICAL METHOD

- Place table at P and level.
- Fix tracing paper, mark point p.
- Sight A, B, C → rays may not pass through a, b, c.
- Move tracing paper so rays pass through a, b, c simultaneously.
- Prick p on map.
- Remove tracing paper and check by drawing rays to A, B, C from p.

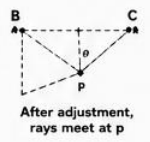


(iii) TRIAL AND ERROR METHOD

- Place table at P and level.
- Orient approximately.
- Draw rays to A, B, C → they form triangle of error.
- Rotate table slightly clockwise / anticlockwise.
- Repeat until rays intersect at p.
- Mark P on ground.



Rays form triangle of error



POINTS TO BE KEPT IN MIND IN PLANE TABLING

- Ground points: A, B, C... and plan points (on paper): a, b, c...
- Rays from survey stations to objects shall be drawn by dashed line.
- Alidade should be properly pivoted while sighting.
- First survey station and scale of the map shall be so chosen that entire area can be plotted on paper.
- While establishing magnetic north using trough compass, keep away objects causing local attraction.
- Table should be clamped after centering and leveling. Rotate table only during orientation.

ERRORS IN PLANE TABLING

1. INSTRUMENTAL ERRORS

- Board surface not plane.
- Edge of alidade not straight.
- Object vane & sight vane not perpendicular.
- Edge of alidade not parallel to line of sight.
- Fixing clamp not proper.

2. ERRORS IN MANIPULATION & SIGHTING

- Defective leveling
- Defective sighting
- Defective orientation
- Defective centering
- Movement of board between sights

3. ERRORS IN PLOTTING

- Defective scale of map
- Wrongly intersecting rays from two different stations

Symbols: A, B, C... = Survey Stations (Ground)

a, b, c... = Points on Sheet

P = Object / Unknown Station

N ↑ N = Magnetic North